

Universal Realisation of Independence Systems by Acyclic Single-Source Unsplittable-Flow Gadgets

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Abstract

Single-source unsplittable-flow rounding asks whether a feasible fractional flow can be replaced by one path per terminal while controlling both arc congestion and cost. We study the combinatorial expressive power hidden inside the additive-congestion constraint. We introduce a path-closed notion of realising an independence system by the zero-cost choices of primary terminals in a directed acyclic flow instance. The definition quantifies over every directed source-terminal path and therefore remains valid under prefix borrowing, suffix splicing, and hybrid routes.

Our first result is universal: every finite loopless independence system has a polynomial-size realisation, measured in the incidence size of its minimal forbidden sets. Hence every finite simple graph, and more generally every hypergraph independence system without singleton forbidden hyperedges, is representable by an acyclic single-source gadget. We then specialise the construction to odd cycles. For C_{2k+1} , a uniform rational family produces a fractional cheap-selection vector that violates the odd-cycle inequality. A potential shift converts a signed connector separator into nonnegative arc costs and gives the exact cost-preserving additive-congestion threshold

$$\tau = 1 - bq.$$

Within the symmetric family, the supremum threshold is $\frac{k+2}{2(k+1)}$, which tends to $1/2$. For C_5 , an exact certificate independently derives all source-terminal paths and enumerates all $3^{10} = 59,049$ unsplittable routings using rational arithmetic.

1 Introduction

In the single-source unsplittable-flow problem, a common source supplies several terminals with unrelated positive demands. A fractional flow may split a terminal’s demand over several paths, whereas an unsplittable routing must choose a single source-terminal path for each demand. The classical theorem of Dinitz, Garg, and Goemans shows that a feasible fractional flow can always be rounded while increasing the load on each arc by at most the maximum demand [1]. A long-standing conjecture of Goemans asks whether the same additive guarantee can be achieved without increasing total cost. Recent work has established related implications and positive results in structured graph classes, while the general cost-preserving statement remains open [2, 3, 4, 5].

This article takes a structural rather than algorithmic viewpoint. Instead of asking how to round an arbitrary instance, we ask which discrete choice systems can be encoded by the additive-congestion inequalities of a single-source instance. A primary terminal is interpreted as “selected” when it uses its unique zero-cost path. Auxiliary terminals enforce incompatibilities through ordinary shared arc loads. The central difficulty is path closure: a valid construction cannot prescribe

a short menu of routes and ignore other paths that the graph itself creates. Any source prefix may combine with any reachable suffix, so a sound proof must survive all borrowed and hybrid routes.

1.1 Contributions

The main contributions are as follows.

1. We define *strong realisation*, which quantifies over every actual directed path in the gadget rather than over a prescribed path decomposition.
2. We give an explicit acyclic construction that strongly realises every finite loopless independence system. Its size is linear in the ground-set size plus the total incidence size of the minimal forbidden sets.
3. We derive a uniform odd-cycle family whose fractional cheap-selection vector violates the stable-set odd-cycle inequality.
4. We construct nonnegative arc costs and determine the exact cost-preserving additive-congestion threshold of the family.
5. We supply an exact C_5 certificate and an all-path, all-routing verifier based solely on rational arithmetic.

1.2 Scope

The odd-cycle family is not a counterexample to the additive- D cost conjecture: its exact threshold is strictly below D and approaches $D/2$. The result is instead an explicit lower-bound mechanism showing that stable-set obstructions can arise inside ordinary flow-load constraints. The optimisation statement in Section 7 is scoped to the uniform symmetric family; it does not rule out stronger unrelated gadgets.

2 Related work

Dinitz, Garg, and Goemans proved that every single-source fractional flow admits an unsplittable routing whose arc load is at most the fractional load plus the largest demand [1]. Morell and Skutella developed a short proof and a lower-bound counterpart, and formulated a stronger simultaneous-deviation perspective [2]. Traub, Vargas Koch, and Zenklusen obtained results for planar and bounded-genus graphs [5]. Almoghrabi, Skutella, and Warode proved strong decomposition results for series-parallel digraphs [3]. Swamy, Traub, Vargas Koch, and Zenklusen showed how an unweighted error-bounded variant implies a relaxed cost guarantee [4].

The present gadgets also touch graph-representation and conflict-flow literatures, but the modelling mechanism is different. EPT and EPG representations define adjacency through intersections among chosen paths [6]. Minimum-cost flow with conflicts adds an external conflict relation on arcs [7]. Here the forbidden system is induced internally by fractional baseline loads, auxiliary demands, and additive arc inequalities. A targeted literature review found no directly equivalent universal realisation theorem, but a priority claim would require a broader expert bibliographic review.

3 Model and definitions

Let $G = (V, A)$ be a finite directed acyclic graph with source s , terminal set T , and positive terminal demands d_t . A feasible fractional flow $x \in \mathbb{R}_{\geq 0}^A$ has divergence $-\sum_{t \in T} d_t$ at s , divergence d_t at

terminal t , and zero divergence at every other vertex. Arc costs are nonnegative and measured per unit of flow: $c \in \mathbb{R}_{\geq 0}^A$. Let

$$D := \max_{t \in T} d_t.$$

An unsplittable routing $P = (P_t)_{t \in T}$ chooses a directed s - t path for each terminal. Its load on an arc a is

$$y_P(a) = \sum_{t: a \in P_t} d_t,$$

and its cost is $c^T y_P$.

Definition 1 (Additive goodness). For $\lambda \geq 0$, a routing P is λ -good if

$$y_P(a) \leq x(a) + \lambda D \quad \text{for every } a \in A.$$

We distinguish *primary* terminals from auxiliary terminals. A primary terminal is *cheaply selected* when its chosen path has total cost zero.

Definition 2 (Strong realisation). Let $\mathcal{I} \subseteq 2^E$ be an independence system on a finite ground set E . A flow instance strongly realises $\mathcal{I}$ when:

1. primary terminals are indexed by E ;
2. every primary terminal has exactly one zero-cost path; and
3. for every $S \subseteq E$, there exists a 1-good routing that cheaply selects exactly S if and only if $S \in \mathcal{I}$.

The quantifier ranges over all directed paths in G .

An independence system is *loopless* when every singleton belongs to it. Equivalently, every inclusion-minimal forbidden set has cardinality at least two. Write $\mathcal{F}$ for the antichain of minimal forbidden sets. Then

$$S \in \mathcal{I} \iff F \not\subseteq S \text{ for every } F \in \mathcal{F}.$$

4 A universal realisation theorem

We now give the construction in explicit vertex-and-arc form. This makes the acyclicity, path closure, and fractional feasibility transparent.

4.1 Construction

Fix $i \in E$, and order the forbidden sets containing i as

$$F_{i,1}, F_{i,2}, \dots, F_{i,m_i}.$$

When $m_i \geq 1$, introduce private vertices $u_{i,j}, v_{i,j}$ for $j = 1, \dots, m_i$. Add:

- a start arc $s \rightarrow u_{i,1}$,
- incidence resources $r_{F_{i,j},i} : u_{i,j} \rightarrow v_{i,j}$,
- connectors $v_{i,j} \rightarrow u_{i,j+1}$ ($j < m_i$),
- a final arc $v_{i,m_i} \rightarrow t_i$,
- a direct arc $s \rightarrow t_i$,
- approaches $s \rightarrow u_{i,j}$ ($j \geq 2$).

If $m_i = 0$, retain a private two-arc chain from s to t_i together with the direct arc.

For each $F \in \mathcal{F}$, create an auxiliary terminal a_F . Whenever $F = F_{i,j}$, add an exit

$$v_{i,j} \rightarrow a_F.$$

Thus any path ending at a_F has a final exit associated with some incidence (F, i) , and immediately before that exit it must traverse $r_{F,i}$.

All chain, incidence, connector, final, and exit arcs have zero cost. Direct arcs and approaches have any strictly positive cost. The graph is acyclic: place s first, order the private chain vertices from left to right, and place all primary and auxiliary terminals last.

4.2 Fractional flow

Set $q = 1/4$, and give every terminal demand 1, so $D = 1$. For each primary i , send q units along its full chain and $1 - q$ along its direct arc. For each auxiliary a_F , split one unit uniformly among the $|F|$ incidences. At a first incidence use the chain start; at a later incidence use its private approach. In either case, traverse only $r_{F,i}$ and then exit to a_F .

The important fractional loads are

$$x(r_{F,i}) = q + \frac{1}{|F|} < 1$$

and

$$x(\text{connector}) = q.$$

All other loads follow directly from the described path flow, so flow conservation holds at every vertex.

Lemma 3 (Exit dominator). *Every directed s - a_F path contains an incidence resource $r_{F,i}$ immediately before its final exit.*

Proof. The only arcs entering a_F are the exits $v_{i,j} \rightarrow a_F$ for incidences with $F_{i,j} = F$. The only arcs entering the corresponding vertex $v_{i,j}$ are the resource arcs $r_{F,i} : u_{i,j} \rightarrow v_{i,j}$. Hence any path using that exit must first traverse $r_{F,i}$. A borrowed prefix may add earlier resources but cannot bypass this last one. $\square$

Lemma 4 (Unique cheap primary path). *For each primary terminal t_i , its full private chain is its unique zero-cost path.*

Proof. The direct arc has positive cost. Any path entering the chain downstream uses a positive-cost approach. There are no arcs from another chain or from an auxiliary terminal into the chain, so the only remaining s - t_i path is the full chain, whose arcs all have zero cost. $\square$

Theorem 5 (Universal realisation). *Every finite loopless independence system has a strong realisation in an acyclic single-source unsplittable-flow instance. The construction has size*

$$O\left(|E| + \sum_{F \in \mathcal{F}} |F|\right).$$

Proof. Lemmas 3 and 4 establish path closure and unique cheap selection.

For the forward direction, suppose a 1-good routing selects every element of some $F \in \mathcal{F}$. The auxiliary a_F must use a final exit associated with some $i \in F$. By Lemma 3, its path traverses $r_{F,i}$. The selected primary i also traverses this resource, so its integral load is 2. However,

$$x(r_{F,i}) + D = q + \frac{1}{|F|} + 1 \leq \frac{1}{4} + \frac{1}{2} + 1 < 2,$$

contradicting 1-goodness. Thus the selected set contains no minimal forbidden set and belongs to $\mathcal{I}$.

Conversely, take $S \in \mathcal{I}$. For every $F \in \mathcal{F}$, choose $i_F \in F \setminus S$. Route each selected primary along its full chain and each unselected primary directly. Route a_F locally through r_{F,i_F} : use the start path when this is the first incidence of chain i_F , and otherwise use the private approach. No auxiliary shares an incidence resource with a selected primary. Every resource load is therefore at most 1, every selected-chain arc has load at most 1, and every direct or approach arc has load at most 1. Since all fractional loads are nonnegative and $D = 1$, every arc load is at most $x + D$. The routing is 1-good and selects exactly S .

There are two private vertices and a constant number of arcs per incidence, plus one terminal and a constant number of arcs per ground element or forbidden set, giving the stated size bound. $\square$

Corollary 6. *Every finite simple graph is strongly realisable by taking its edges as the minimal forbidden sets. More generally, every finite hypergraph independence system with no singleton forbidden edge is strongly realisable.*

This includes complete graphs, odd holes, odd antihole, webs, antiwebs, and minimally imperfect graphs. The point is not that these graph classes share a path-intersection representation, but that their stable sets can be encoded by ordinary load inequalities in a single-source flow instance.

5 A uniform odd-cycle family

Fix $k \geq 1$, put $n = 2k + 1$, and index vertices modulo n . Let $e_i = \{i, i + 1\}$. On primary chain i , order the incidence for e_{i-1} first, insert a connector h_i , and place the incidence for e_i second. Add a private approach from s to the second incidence.

Choose rational

$$0 < \varepsilon < \frac{1}{4(k+1)}$$

and define

$$T = \frac{k}{2(k+1)}, \quad t = T + \varepsilon, \quad s_0 = \frac{1}{2} + \varepsilon,$$

$$b = s_0 + t, \quad q = \frac{t}{b}.$$

Primary demands are b , edge-auxiliary demands are 1, and hence $D = 1$. Send $t = bq$ units of each primary along its full chain and $s_0 = b(1 - q)$ along its direct arc. Split each edge-auxiliary flow equally between the first incidence at $i + 1$ and the approached second incidence at i . Every incidence resource has fractional load $t + 1/2$, and every connector has fractional load t .

Theorem 7 (Exact stable-set correspondence). *The 1-good cheap selections of the odd-cycle instance are exactly the stable sets of C_n .*

Proof. If adjacent primaries i and $i + 1$ are both selected, the edge auxiliary for e_i must traverse an incidence resource of one endpoint. That resource has load $b + 1$, whereas its 1-good allowance is

$$t + \frac{1}{2} + 1.$$

The excess is

$$b + 1 - \left(t + \frac{1}{2} + 1\right) = s_0 - \frac{1}{2} = \varepsilon > 0.$$

Thus a selected set is stable.

Conversely, let S be stable. For each edge, route its auxiliary through an unselected endpoint, using the private approach when that endpoint's incidence is second. No selected primary shares an incidence resource with an auxiliary. Direct inspection of starts, connectors, approaches, exits, and direct arcs shows that every load is at most its fractional load plus D . Hence the routing is 1-good and selects exactly S . $\square$

The fractional cheap-selection coordinate of every primary is q . Since

$$nq - k = \frac{\varepsilon}{b} > 0, \tag{1}$$

the vector $(q, \dots, q)$ violates the odd-cycle inequality

$$\sum_{i=0}^{n-1} z_i \leq k.$$

6 Nonnegative costs and an exact threshold

To state the result precisely, define the cost-preserving additive-congestion threshold of an instance by

$$\tau^* := \inf \left\{ \lambda \geq 0 : \begin{array}{l} \text{there exists a } \lambda\text{-good unsplittable routing } P \\ \text{with } c^T y_P \leq c^T x \end{array} \right\}.$$

Start with a signed weight p that assigns $-1/b$ to every connector h_i and 0 to every other arc. Give potential 0 to vertices before a connector and $-1/b$ to vertices after it, including primary and auxiliary terminals. Define reduced costs

$$c(u, v) = p(u, v) + \pi(u) - \pi(v).$$

These costs are nonnegative. Equivalently, cost $1/b$ is placed on each direct primary arc, each approach to a second incidence, and each exit from a first incidence; every other arc has cost zero.

Fractional and integral routings have identical divergence. Consequently the potential terms cancel:

$$c^T(y - x) = p^T(y - x). \tag{2}$$

Theorem 8 (Exact cost-congestion threshold). *For the uniform odd-cycle family,*

$$\tau^* = 1 - t = 1 - bq.$$

More precisely, for every $\lambda < 1 - t$ and every λ -good unsplittable routing y ,

$$c^T y \geq c^T x + nq - k = c^T x + \frac{\varepsilon}{b}.$$

At $\lambda = 1 - t$, there is an unsplittable routing of cost at most $c^T x$.

Proof. A borrowed auxiliary route to a second incidence uses connector h_i and places demand 1 on it. Because $x(h_i) = t$, such a route is infeasible whenever

$$1 > t + \lambda,$$

that is, whenever $\lambda < 1 - t$. In this range, connector h_i has integral load b exactly when primary i is selected, and zero otherwise.

If two adjacent primaries were selected, their edge auxiliary could not borrow a connector and would have to share an incidence resource with one endpoint. Its load would be $b + 1$, while

$$t + \frac{1}{2} + \lambda < t + \frac{1}{2} + 1 - t = \frac{3}{2} < b + 1.$$

Hence the selected set is stable and has cardinality at most k . Therefore

$$p^T y \geq -k, \quad p^T x = -nq.$$

Equation (2) and (1) give

$$c^T(y - x) = p^T(y - x) \geq nq - k = \frac{\varepsilon}{b}.$$

At $\lambda = 1 - t$, route every primary directly. Route each edge auxiliary along the full borrowed prefix of one primary chain to its second incidence. Every connector then carries exactly

$$1 = t + (1 - t),$$

and all other bounds hold because $t < 1/2$. The routing costs n , while

$$c^T x = n \left(1 - q + \frac{1}{b} \right) > n.$$

Thus a cost-preserving routing first appears at $\lambda = 1 - t$. □

7 Optimisation within the symmetric family

Write $t = bq$ and $s_0 = b(1 - q)$. Exact stable-set realisation requires $s_0 > 1/2$, while violation of the odd-cycle inequality requires $nq > k$. Since $n = 2k + 1$,

$$(k + 1)t > ks_0 > \frac{k}{2}.$$

It follows that

$$\tau^* = 1 - t < \frac{k + 2}{2(k + 1)}.$$

Our ε -family attains

$$\tau^* = \frac{k + 2}{2(k + 1)} - \varepsilon.$$

Therefore the exact supremum within this uniform symmetric parameterisation is

$$\tau_k^{\sup} = \frac{k + 2}{2(k + 1)} = \frac{1}{2} + \frac{1}{2(k + 1)}, \quad \lim_{k \rightarrow \infty} \tau_k^{\sup} = \frac{1}{2}.$$

Table 1: Exact parameters of the C_5 certificate.

b	q	$t = bq$	τ^*	$5q - 2$	$c^T x$
$\frac{17}{20}$	$\frac{41}{102}$	$\frac{41}{120}$	$\frac{79}{120}$	$\frac{1}{102}$	$\frac{905}{102}$

8 An exact C_5 certificate

For $k = 2$, choose $\varepsilon = 1/120$. Table 1 records the resulting parameters.

The machine-readable DAG has 31 vertices and 45 arcs. Every one of the five primary and five auxiliary terminals has exactly three actual paths, so the verifier enumerates

$$3^{10} = 59,049$$

unsplittable routings. It derives the path sets from the arc list and does not trust a prescribed decomposition.

The independent verifier establishes:

- fractional feasibility and acyclicity;
- exactly one zero-cost path for each primary terminal;
- exactly the 11 stable sets of C_5 as cheap masks at $\lambda = 1$;
- at $\lambda = 7/12$, minimum integral cost $151/17$, exceeding $c^T x = 905/102$ by exactly $1/102$;
- at $\tau^* = 79/120$, a routing of cost $5 < c^T x$;
- rejection of mutations that remove an incompatibility, add a bypass, create a cheap hybrid primary route, or erase the odd-cycle violation.

All arithmetic uses Python’s exact `Fraction` type. The finite certificate supports the general proof but does not replace it: exhaustive enumeration is restricted to C_5 , while the universal and odd-cycle theorems are symbolic.

9 Discussion and limitations

The universal construction shows that additive load inequalities can encode arbitrary finite independence systems, provided singleton selection remains feasible. This expressive power comes from three ingredients: a private zero-cost chain for each decision, an incidence resource for each participation in a minimal obstruction, and an auxiliary unit demand for each obstruction. The exit-dominator property makes the mechanism robust to path splicing.

The construction size is polynomial in an explicit minimal-forbidden-set description, not necessarily in a more compressed oracle or graph representation. An independence system can have exponentially many minimal forbidden sets, and the theorem does not avoid that representation cost.

The exact threshold is proved only for the uniform symmetric odd-cycle family. It does not imply that $1/2$ is a universal barrier, nor does it refute the additive- D cost conjecture. Finally, the general proof has not been formalised in a proof assistant. The exact scripts machine-check the finite certificate and rational identities, while the universal theorem remains a conventional mathematical proof.

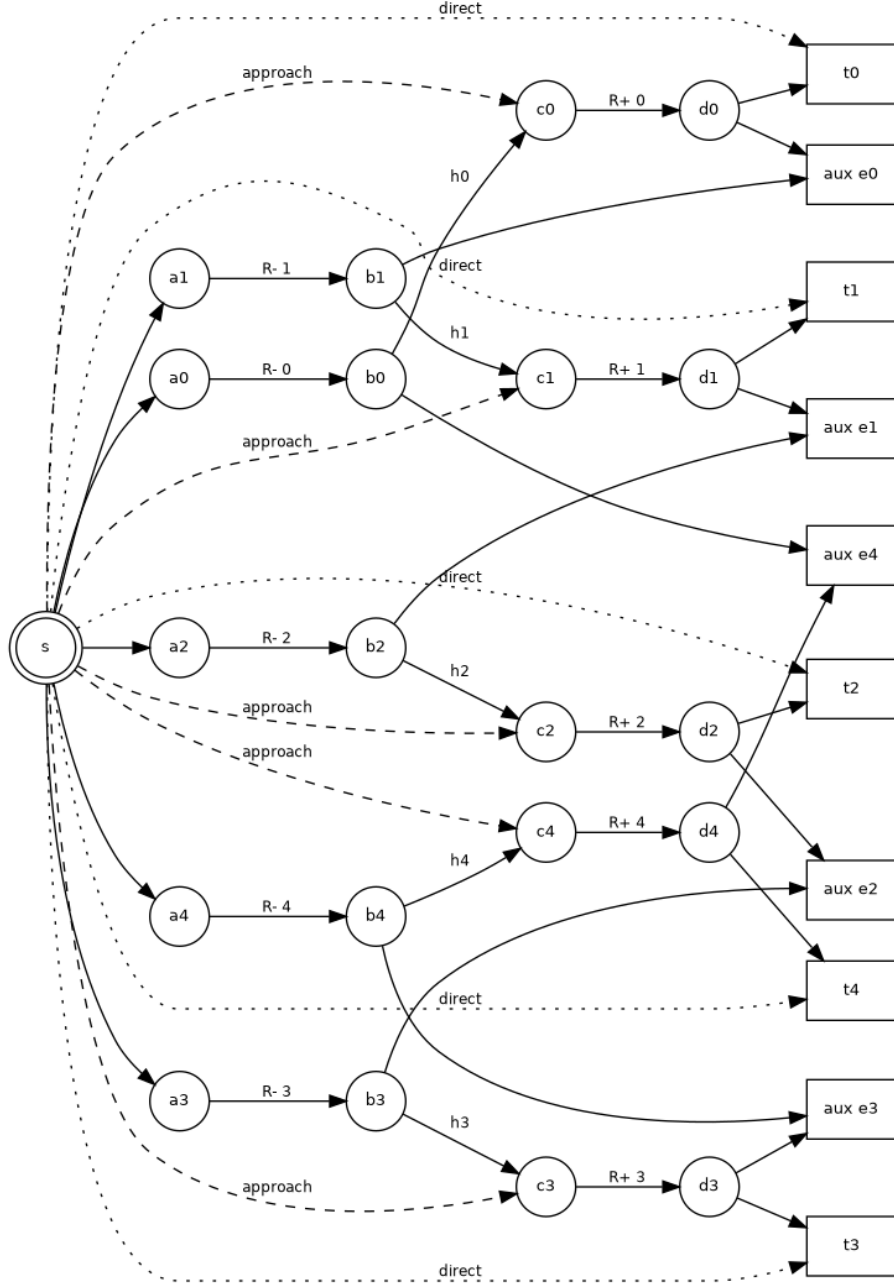


Figure 1: The exact C_5 DAG. Solid arcs are zero-cost chain, resource, connector, or exit arcs; dashed approaches and dotted direct arcs carry positive cost. Labels R_i^- , h_i , and R_i^+ denote the first resource, connector, and second resource of primary chain i .

10 Conclusion

Every finite loopless independence system can be embedded into the cheap-choice structure of an acyclic single-source unsplittable-flow instance in a manner closed under all actual graph paths. Odd cycles then turn this qualitative universality into a quantitative cost-congestion obstruction: their fractional cheap-selection vectors violate the stable-set inequality, and a potential shift yields explicit nonnegative costs with an exact threshold. The C_5 certificate demonstrates that the phenomenon is fully reproducible at finite scale.

The construction suggests two natural directions. First, one may seek compressed gadgets for structured forbidden systems. Second, one may investigate which additional graph restrictions—planarity, bounded genus, bounded treewidth, or series-parallel structure—limit the independence systems that can be realised and thereby constrain cost-congestion lower bounds.

A Reproducibility protocol

The accompanying package contains the C_5 JSON instance, the exhaustive verifier, and the symbolic parameter checker. From the package directory, run

```
python3 verify_c5.py C5.json
python3 verify_symbolic.py --max-k 200
```

The first command enumerates every path and routing of the finite certificate. The second checks the parameter identities and inequalities for $k = 1, \dots, 200$. Neither command requires a third-party Python package.

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